

⚠ DRAFT v3.0 — proposed upgrade, NOT clinically validated.

Every new or changed threshold below is tagged **DRAFT · SIGN-OFF** and must be approved by the AIH&PH / WB Health Secretariat clinical committee before it touches `clinical_decision_engine.json` or `triage_cases.json`.

AshaMitra — Clinical Decision Rules v3.0 (DRAFT for clinical sign-off)

Status: Proposed upgrade to engine `asha_cdss_engine v1.0.0` / cases `v2.0.0`. **Author of this draft:** Tooling/design proposal — **NOT a clinical authority. Nothing in this document is validated.** Every new or changed clinical threshold below is tagged **[DRAFT – SIGN-OFF]** and must be reviewed and approved by **AIH&PH / WB Health Secretariat clinical committee** before it touches `clinical_decision_engine.json` or `triage_cases.json`.

Design intent. This upgrade does **not** try to make the app "diagnose like a doctor." The ASHA's role is *recognise danger signs → refer correctly*. The clinical expertise lives in the **rules** (committee-validated); the engine stays deterministic, offline, no-LLM, and biased toward over-referral. The goal of v3.0 is **completeness, correct thresholds, and safe handling of uncertainty** — so the tool remains defensible under review.

Protocol basis (unchanged): HBNC, IMNCI (incl. Young Infant 0–2 m), HBYC, MCP, PMSMA, SBA, PPH, PNC, UIP.

PART A — Design-layer upgrades (apply across all modules)

These are higher-impact than any single rule, because they fix gaps that affect *every* case.

A1. Universal danger-sign pre-sweep [DRAFT – SIGN-OFF]

Problem: IMNCI opens every child assessment with four general danger signs. The current cases jump straight into module-specific questions, so a generic danger sign can be missed if it doesn't match a module question.

Fix: Add a mandatory **pre-sweep** that runs *before* the module questions for any paediatric case (Newborn / Infant / Child). Any `হ্যাঁ` → immediate RED, band-locked, skip to referral.

Pre-sweep ID	Question (BN / EN)	Answers	Logic
GDS-01	শিশু কি কিছু খেতে বা বুকের দুধ খেতে পারছে না? / Unable to drink or breastfeed at all?	হ্যাঁ · না · নিশ্চিত না	হ্যাঁ → RED
GDS-02	যা-ই খাচ্ছে সব বমি করে ফেলছে? / Vomits everything?	হ্যাঁ · না · নিশ্চিত না	হ্যাঁ → RED
GDS-03	খিঁচুনি হচ্ছে বা হয়েছে? / Convulsions now or recently?	হ্যাঁ · না · নিশ্চিত না	হ্যাঁ → RED
GDS-04	নিশ্বেজ, অজ্ঞান, বা ডাকে সাড়া দিচ্ছে না? / Lethargic, unconscious, or not responding?	হ্যাঁ · না · নিশ্চিত না	হ্যাঁ → RED

Referral: FRU / SNCU / DH immediately, call 108. (For a young infant 0–2 m, "unable to feed" includes "feeding very little / stopped sucking.")

A2. Uncertainty policy — stop "নিশ্চিত না → GREEN" [DRAFT – SIGN-OFF]

Problem: Every rule's logic tests only = হ্যাঁ. A নিশ্চিত না (unsure) on a hard-stop danger-sign question fires nothing and can land the patient in GREEN. A clinician resolves uncertainty by referring, not by clearing.

Proposed rule (committee to pick the exact policy):

- On any **hard-stop** question, নিশ্চিত না → re-prompt once with a plain re-check instruction ("look again / ask the family"). If still নিশ্চিত না :
 - Option 1 (recommended, safer):** force minimum **YELLOW** (PHC within 24 h) and **block GREEN** for the whole case.
 - Option 2 (less referral load):** treat as না but **block GREEN** — case can be at most "watch + follow-up in 2 days," never "all clear."
- Trade-off:** Option 1 raises YELLOW referral volume at PHC but closes the silent-miss path; Option 2 keeps volume lower but relies on follow-up. Pick per PHC capacity. Either way, **GREEN must require all danger-sign answers to be a clear না**.

A3. Score every answer option, not just হ্যাঁ [DRAFT – SIGN-OFF]

Problem: Answer sets offer intermediate options (খুব কম , কিছুটা , একটু , অনেক , মাঝে মাঝে , একবার) but per-rule logic ignores them. e.g. NB-001 offers "খুব কম" (feeding very little) — clinically PSBI — yet logic only fires on হ্যাঁ .

Fix: Each rule must declare an explicit **answer→band map**. Defaults to standardise:

Option	Meaning	Default band on a danger-sign question
হ্যাঁ	Yes	as the rule specifies (usually RED)
খুব কম / অনেক / একটু	severe-degree variants	same band as হ্যাঁ (a degree of the danger sign)
একবার	once (e.g. one convulsion/faint)	RED for convulsion/LOC/bleeding questions
মাঝে মাঝে / কিছুটা	intermittent / partial	YELLOW (not GREEN)
না	No	no fire
নিশ্চিত না	Unsure	per A2

Per-rule overrides are listed in the rule cards where they differ.

A4. Structured measurement capture [DRAFT – SIGN-OFF, hardware-dependent]

Problem: RR count, MUAC, temperature, BP, Hb, pad-soak, fetal kick-count are referenced in *actions* but never captured as numeric inputs that drive the band — so a yes/no proxy stands in for a measurable fact. Numbers + thresholds are what make triage precise.

Fix: Where the ASHA has the device, add a numeric input that the engine thresholds deterministically. **Confirm device availability for your WB ASHA cohort before enabling** — if a device isn't in the kit, keep the yes/no proxy and don't ask for the number.

Input	Device	Deterministic thresholds (to confirm at sign-off)
Respiratory rate /min	timer/phone	newborn/young-infant ≥ 60 RED; 2–12 m ≥ 50 ; 12 m–5 y ≥ 40 → pneumonia
Temperature °C	digital thermometer	newborn/young-infant: ≥ 37.5 or < 35.5 → RED
MUAC cm	MUAC tape	< 11.5 → SAM/RED; 11.5–12.5 → MAM/YELLOW
BP mmHg	BP machine (if in kit)	$\geq 140/90$ → RED (pre-eclampsia)
Hb g/dL	Hb meter (if available)	< 7 → RED; 7–10.9 → YELLOW (anaemia)
Pad soak	count + time	≥ 2 pads soaked in 30 min → RED *(proxy still pending per README item #3)*

A5. Per-case age / gestational-age intake [DRAFT – SIGN-OFF]

Problem: Several rules reference age-dependent thresholds ("infant < 2 m", "fever < 3 m", jaundice timing) but no question captures exact age, and bleeding in pregnancy is interpreted without trimester.

Fix:

- **Newborn/Infant:** capture **age in days/weeks** at intake. Drive young-infant (0–2 m) vs older-infant (2–12 m) logic from it (feeding refusal, any fever, and any low temp are RED in 0–2 m).
- **Pregnancy:** capture **gestational age / trimester**. Bleeding 1st trimester → miscarriage/ectopic pathway; bleeding ≥ 20 weeks → placenta praevia/abruption — both RED, but the referral note and "no vaginal exam" warning differ.

PART B — Per-module rule changes & additions

Existing rules not listed here are **retained unchanged**. Only corrections and [NEW] rules are shown.

1. 🧒 Newborn Checkup (0–28 days) — newborn (HBNC + IMNCI Young Infant)

Corrections to existing rules:

- **NB-001** — add answer-map: খুব কম (feeding very little) → **RED** (PSBI "very poor feeding"), not silent.
- **NB-006 (jaundice/cyanosis)** — **split the conflated logic** [DRAFT – SIGN-OFF]. Current logic fires RED on any হ্যাঁ but the note describes a YELLOW day-2–14 path. Replace with a follow-up:
 - n6a: ত্বক/ঠোঁট/জিভ নীলাভ? / bluish skin, lips, or tongue (cyanosis)? → **RED always**.
 - n6b: জন্মের ২৪ ঘণ্টার মধ্যে হলুদ হয়েছে, বা হাতের তালু/পায়ের তলা হলুদ? / jaundice within 24 h of birth **or** palms/soles yellow? → **RED**.
 - n6c: হলুদভাব ২–১৪ দিনে, হাতের তালু/পায়ের তলা স্বাভাবিক? / jaundice day 2–14, palms/soles spared? → **YELLOW** (PHC within 24 h for bilirubin).

New rules:

NB-008 RED HARD-STOP n8 → SNCU / FRU immediately [NEW] [DRAFT – SIGN-OFF]

শিশুর শরীর কি ঠান্ডা লাগছে বা স্বাভাবিকের চেয়ে অনেক কম গরম?

Is the baby cold to touch / colder than normal (hypothermia)?

ANSWERS হ্যাঁ · না · নিশ্চিত না

LOGIC IF n8 = হ্যাঁ THEN RED. If thermometer present and temp < 35.5°C THEN RED.

ACTION PSBI hard-stop. Skin-to-skin/kangaroo warmth + cover head, keep warm during transport.

Refer SNCU/FRU immediately (≤30 min). Hypothermia is a major neonatal killer.

MAPS TO cold to touch · low body temperature · temp < 35.5°C · hypothermia

NB-009 RED HARD-STOP n9 → SNCU / FRU immediately [NEW] [DRAFT – SIGN-OFF]

শিশুর খঁচুনি হচ্ছে, হাত-পা শক্ত হয়ে যাচ্ছে বা পিছনে বেঁকে যাচ্ছে?

Convulsions, stiffening, or back-arching (opisthotonus)?

ANSWERS হ্যাঁ · না · একবার · নিশ্চিত না

LOGIC IF n9 IN [হ্যাঁ, একবার] THEN RED.

ACTION PSBI hard-stop. Protect airway, left-lateral, do not restrain limbs. Call 108. SNCU/FRU now.

MAPS TO neonatal seizure · convulsion · stiffening · arching · twitching

NB-010 RED HARD-STOP n10 → FRU / DH [NEW] [DRAFT – SIGN-OFF]

ত্বকে অনেক ফুসকুড়ি/পুঁজভরা ফোঁসা, অথবা কান থেকে পুঁজ পড়ছে?

Many skin pustules / pus-filled blisters, or pus draining from ear?

ANSWERS হ্যাঁ · না · অল্প কয়েকটা · নিশ্চিত না

LOGIC IF n10 = হ্যাঁ THEN RED. IF n10 = অল্প কয়েকটা (few localised pustules) THEN YELLOW.

ACTION Many/severe pustules or ear pus = PSBI local infection → FRU/DH. Few localised = PHC within 24 h.

MAPS TO skin pustules · pus-filled blisters · ear discharge · local bacterial infection

NB-011 RED HARD-STOP n11 → SNCU / FRU immediately [NEW] [DRAFT – SIGN-OFF]

মাথার নরম অংশ (তালু) ফুলে উঁচু হয়ে আছে?

Is the soft spot on the head (fontanelle) bulging?

ANSWERS হ্যাঁ · না · নিশ্চিত না

LOGIC IF n11 = হ্যাঁ THEN RED.

ACTION Possible meningitis. Refer SNCU/FRU immediately.

MAPS TO bulging fontanelle · raised intracranial pressure · meningitis sign

Optional intake flags for HBNC counselling (not RED by themselves): **LBW / preterm** (birth weight <2.5 kg or born early) and **did-not-cry-at-birth** history → mark for closer follow-up + extra warmth/ feeding counselling. Confirm whether these should escalate band at sign-off.

Newborn band tally after upgrade: RED 9–10, YELLOW (jaundice split / few pustules) 1–2, GREEN 1.

2. 🧒 Child Health Check (1–5 years) — child (IMNCI + HBYC)

Corrections:

- **CH-002 (cough/breathing)** — make the escalation deterministic [DRAFT – SIGN-OFF] : add explicit follow-ups instead of prose. c2a: বুক ভিতরের দিকে টেনে বসছে (chest indrawing)? → **RED**. c2b: শ্বাসে শোঁ-শোঁ আওয়াজ/স্ট্রিডার? → **RED**. RR input (A4): $\geq 40/\text{min}$ (1–5 y) → **YELLOW** pneumonia.
- **CH-003 (diarrhoea)** → **add dehydration sub-assessment** [DRAFT – SIGN-OFF] , so the GREEN/YELLOW/RED paths described in the action actually exist as logic:
 - c3a: চোখ গর্তে বসে গেছে? / sunken eyes? · c3b: চামড়া টেনে ছাড়লে ধীরে ফিরছে? / skin pinch slow? · c3c: পান করতে পারছে না বা নিস্তেজ? / unable to drink or lethargic?
 - Any of c3a–c3c **হ্যাঁ** → **RED (severe dehydration)**. Restless/irritable + drinks eagerly + some signs → **YELLOW (some dehydration)**. None → **GREEN (ORS at home)**.
 - c3d: পায়খানা ১৪ দিনের বেশি? / diarrhoea ≥ 14 days (persistent)? → **YELLOW**, PHC.

New rules:

CH-010 RED HARD-STOP c7 → FRU / DH immediately [NEW] [DRAFT – SIGN-OFF]
ঘাড় শক্ত, আলোয় কষ্ট, বা প্রচণ্ড মাথাব্যথা (জ্বরসহ)?
Stiff neck, light sensitivity, or severe headache (with fever)?
ANSWERS হ্যাঁ · না · নিশ্চিত না
LOGIC IF c7 = হ্যাঁ THEN RED.
ACTION Possible meningitis/encephalitis. Refer FRU/DH immediately, call 108.
MAPS TO neck stiffness · photophobia · meningitis · encephalitis · severe headache with fever

CH-011 YELLOW SCORED c8 → PHC within 24 h (RED if signs of shock) [NEW] [DRAFT – SIGN-OFF]
পায়খানার সাথে রক্ত যাচ্ছে?
Is there blood in the stool (dysentery)?
ANSWERS হ্যাঁ · না · নিশ্চিত না
LOGIC IF c8 = হ্যাঁ THEN YELLOW (dysentery → antibiotic at PHC). Escalate RED if severe dehydration (link to c3a–c3c) or lethargy.
ACTION Dysentery needs antibiotics – refer PHC within 24 h. With dehydration/lethargy → FRU.
MAPS TO blood in stool · dysentery · mucoid bloody diarrhoea

CH-012 YELLOW SCORED c9 → PHC within 24 h (RED if severe pallor) [NEW] [DRAFT – SIGN-OFF]
হাতের তালু/চোখের পাতা খুব ফ্যাকাশে?
Are palms / inner eyelids very pale (anaemia)?
ANSWERS হ্যাঁ · না · নিশ্চিত না
LOGIC IF c9 = হ্যাঁ THEN YELLOW. Severe palmar pallor → RED.
ACTION Some pallor → PHC for Hb. Severe pallor → FRU (possible transfusion / severe anaemia).
MAPS TO pallor · palmar pallor · conjunctival pallor · anaemia

Measles cross-check (optional YELLOW): generalised rash + fever + cough/red eyes → flag for measles, check Vit-A and immunisation status; corneal clouding or mouth ulcers → RED. Confirm at sign-off.

3. 🧒 Infant Checkup (1–12 months) — `child` (IMNCI + HBYC)

This module has the most dangerous structural gaps. Two fixes are mandatory.

Corrections:

- **Add age-in-weeks intake (A5)** and split **young infant (0–2 m)** vs **older infant (2–12 m)**:
 - 0–2 m: **any** fever, **any** low temp, **any** feeding refusal → **RED** (PSBI). This makes CH-007-i and CH-007 age-correct instead of conflating duration with age.
- **CH-008 (breathing)** — same deterministic split as CH-002: chest indrawing/stridor → RED; RR ≥ 50/min (2–12 m) alone → YELLOW.

New rules (close the silent-miss holes):

CH-013-i RED HARD-STOP i7 → SNCU / FRU immediately [NEW] [DRAFT – SIGN-OFF]
শিশু অনেক কম নড়ছে, নিশ্বেজ, বা ডাকে/স্পর্শে সাড়া দিচ্ছে না?
Is the infant very lethargic, limp, or not responding to voice/touch?
ANSWERS হ্যাঁ · না · কিছুটা · নিশ্চিত না
LOGIC IF i7 = হ্যাঁ THEN RED. IF i7 = কিছুটা (moves only when stimulated) THEN RED in 0–2 m, else YELLOW.
ACTION Lethargy = severe systemic illness. Refer SNCU/FRU immediately.
MAPS TO lethargic infant · limp · unresponsive · moves only when stimulated ⚠ THIS RULE DID NOT EXIST

CH-014-i RED HARD-STOP i8 → SNCU / FRU immediately [NEW] [DRAFT – SIGN-OFF]
শিশুর ঝাঁচুনি হচ্ছে বা হয়েছে?
Convulsions now or recently?
ANSWERS হ্যাঁ · না · একবার · নিশ্চিত না
LOGIC IF i8 IN [হ্যাঁ, একবার] THEN RED.
ACTION Protect airway, left-lateral, call 108, SNCU/FRU now.
MAPS TO infant seizure · convulsion · fits

CH-015-i RED HARD-STOP i9 → SNCU / FRU immediately [NEW] [DRAFT – SIGN-OFF]
শরীর ঠান্ডা (০–২ মাসে) বা মাথার তালু ফুলে উঠে?
Cold body (in 0–2 m) or bulging fontanelle?
ANSWERS হ্যাঁ · না · নিশ্চিত না
LOGIC IF i9 = হ্যাঁ THEN RED. (Hypothermia in young infant; bulging fontanelle = meningitis.)
ACTION Warmth + refer SNCU/FRU immediately.
MAPS TO hypothermia young infant · bulging fontanelle · meningitis sign

Infant band tally after upgrade: RED 7–8, YELLOW 2–3, plus pre-sweep (A1) in front.

4. 🤰 Pregnant Mother Checkup — `pregnancy` (MCP + PMSMA)

Corrections:

- **ANC-001** — separate the two collapsed concepts so the engine reflects them [DRAFT – SIGN-OFF] : p1a BP input (A4) ≥ 140/90 → RED; p1b severe headache → RED; p1c **epigastric / right-upper-abdomen pain** (severe pre-eclampsia / HELLP) → **RED** (currently not captured at all).
- **ANC-002 / ANC-006** escalations retained; ensure they are expressed as `condition_sets`, not prose.

New rules:

ANC-008 RED HARD-STOP p8 → FRU / DH same day (immediately if fainting) [NEW] [DRAFT - SIGN-OFF]

খুব ফ্যাকাশে, দুর্বল, বা সামান্য পরিশ্রমেই হাঁপিয়ে যাচ্ছেন?

Very pale, weak, or breathless on slight exertion (severe anaemia)?

ANSWERS হ্যাঁ · না · কিছুটা · নিশ্চিত না

LOGIC IF p8 = হ্যাঁ THEN RED. IF Hb available and <7 THEN RED; 7-10.9 THEN YELLOW.

IF p8 = কিছুটা (mild pallor, no breathlessness) THEN YELLOW.

ACTION Severe anaemia is a leading maternal killer. Severe pallor/breathlessness → FRU/DH for Hb + transfusion planning. Mild → PHC for Hb + IFA.

MAPS TO severe pallor · breathlessness on exertion · palpitations · severe anaemia in pregnancy

ANC-009 RED HARD-STOP p9 → FRU / DH immediately [NEW] [DRAFT - SIGN-OFF]

জ্বর, বিশেষ করে কাঁপুনি দিয়ে জ্বর আসছে?

Fever, especially with chills/rigors?

ANSWERS হ্যাঁ · না · মাঝে মাঝে · নিশ্চিত না

LOGIC IF p9 = হ্যাঁ THEN RED (sepsis/malaria/chorioamnionitis risk in pregnancy).

IF p9 = মাঝে মাঝে (low-grade, no other sign) THEN YELLOW, PHC within 24 h.

ACTION High/rigor fever in pregnancy → FRU/DH (rule out malaria, UTI, chorioamnionitis, sepsis).

MAPS TO fever in pregnancy · rigors · chills · maternal sepsis · malaria

ANC-010 RED HARD-STOP p10 → FRU / DH immediately [NEW] [DRAFT - SIGN-OFF]

যোনিপথে হঠাৎ জল ভেঙেছে বা ক্রমাগত জল বারছে?

Sudden gush or continuous leaking of fluid (PROM)?

ANSWERS হ্যাঁ · না · নিশ্চিত না

LOGIC IF p10 = হ্যাঁ THEN RED. (Preterm: even more urgent.)

ACTION PROM → infection/preterm-labour risk. No vaginal exam, keep clean pad, refer FRU/DH.

MAPS TO leaking liquor · rupture of membranes · PROM · watery vaginal discharge

Optional intake — High-Risk Pregnancy (HRP) flags (for PMSMA tagging, not necessarily RED alone): age <18 or >35, height <145 cm, previous C-section/stillbirth, multiple pregnancy, malpresentation, known diabetes/hypertension. Decide band at sign-off; minimum is **YELLOW + PMSMA flag**. Note: convulsions/fits in pregnancy are caught by Emergency EM-002 and ANC-006 prodrome.

Pregnancy band tally after upgrade: RED 7, YELLOW 2-3, GREEN 1.

5. 🧑🏻 Postpartum Checkup — **delivery_pnc** (SBA + PPH + PNC)

Biggest single gap: no eclampsia rule. Postpartum (pre-)eclampsia occurs up to ~6 weeks after delivery.

New rules:

PNC-007 RED HARD-STOP pp7 → FRU / DH immediately [NEW] [DRAFT - SIGN-OFF]

প্রসবের পর মাথাব্যথা, চোখে বাপসা দেখা, ঝাঁচুনি, বা রক্তচাপ বেশি?

After delivery: severe headache, blurred vision, convulsions, or high BP?

ANSWERS হ্যাঁ · না · একবার · নিশ্চিত না

LOGIC IF pp7 IN [হ্যাঁ, একবার] THEN RED. IF BP available ≥140/90 THEN RED.

ACTION Postpartum (pre-)eclampsia hard-stop – can occur up to 6 weeks postpartum. Left-lateral, protect airway, call 108, FRU/DH immediately. If convulsing & within ANM/MO scope: MgSO4.

MAPS TO postpartum headache · blurred vision · postpartum convulsion · high BP postpartum · eclampsia

⚠️ THIS RULE DID NOT EXIST – critical addition

PNC-008 RED HARD-STOP pp8 → FRU / DH immediately

[NEW] [DRAFT – SIGN-OFF]

হঠাৎ শ্বাসকষ্ট, বুকে ব্যথা, বা এক পায়ে ব্যথা/ফোলা?

Sudden breathlessness, chest pain, or pain/swelling in one calf?

ANSWERS হ্যাঁ · না · নিশ্চিত না

LOGIC IF pp8 = হ্যাঁ THEN RED. (Pulmonary embolism / DVT / peripartum cardiomyopathy.)

ACTION Refer FRU/DH immediately, call 108. Do not let her walk.

MAPS TO postpartum breathlessness · chest pain · calf swelling · DVT · pulmonary embolism · PPCM

PNC-009 YELLOW SCORED pp9 → PHC / counselling (RED if self-harm thoughts) [NEW] [DRAFT – SIGN-OFF]

মন খুব খারাপ, কাঁদা, ঘুম/খাওয়া কমে যাওয়া, বা বাচ্চার প্রতি আগ্রহ নেই?

Persistent low mood, crying, sleep/appetite loss, or no interest in the baby?

ANSWERS হ্যাঁ · না · মাঝে মাঝে · নিশ্চিত না

LOGIC IF pp9 = হ্যাঁ THEN YELLOW (postpartum depression screen). Any thought of self-harm or harming baby → RED, urgent referral.

ACTION Counsel, involve family, refer PHC/MO. Self-harm risk → escalate immediately.

MAPS TO postpartum depression · low mood · anhedonia · self-harm ideation

Correction — PNC-001 / PNC-002 retained. Keep the [SIGN-OFF PENDING] flag on the "≥2 pads in 30 min" proxy (README item #3). Also add a retained-products/urinary-retention check: **না**-passing urine or palpable bladder → YELLOW (FRU if anuria).

Postpartum band tally after upgrade: RED 3, YELLOW 6–7.

6. 📌 Immunization Missed — immunisation (UIP National Schedule)

The schedule content is out of date. Corrections (verified against current UIP):

- **"MMR" → "MR"** everywhere. India's UIP uses **Measles–Rubella (MR)**, not MMR (no mumps). MR-1 at **9–12 months**, MR-2 at **16–24 months**.
- **Add PCV** (Pneumococcal Conjugate): 2 primary doses at **6 & 14 weeks** + **booster at 9–12 months**.
- **Add Rotavirus (RVV)**: 3 doses at **6, 10, 14 weeks** (oral).
- **Correct IPV → fIPV**: **two fractional doses at 6 & 14 weeks** (not a single IPV).
- **Vitamin A**: 1st dose at **9 months**; doses 2–9 **six-monthly up to 5 years**.
- **PCV/RVV rollout**: now part of national UIP; **confirm current West Bengal rollout/stock status** with the State Immunization Officer before wording catch-up advice (district variation possible).

Corrected reference schedule (for IMM-002 / catch-up logic):

Age	Vaccines (current UIP)
Birth	BCG, OPV-0, Hep-B birth dose
6 weeks	OPV-1, Pentavalent-1, RVV-1, fIPV-1, PCV-1
10 weeks	OPV-2, Pentavalent-2, RVV-2
14 weeks	OPV-3, Pentavalent-3, RVV-3, fIPV-2, PCV-2
9–12 months	MR-1, JE-1 (endemic), PCV-booster, Vitamin A-1
16–24 months	MR-2, JE-2 (endemic), DPT-booster-1, OPV-booster, Vitamin A 2-9 (6-monthly)
5–6 years	DPT-booster-2
10 & 16 years	TT/Td
Pregnant woman	TT/Td (1 dose if vaccinated within 3 years)

Rule logic retained, with one fix and one addition:

- **IMM-001 logic gap:** currently `IF im1 IN [0–6 m, 6–12 m] THEN YELLOW` — make the 1–5 y branch explicitly route to IMM-003 (booster catch-up) so older children aren't dropped.
- **New IMM-006 [NEW] :** AEFI history — "আগের টিকার পর কি গুরুতর প্রতিক্রিয়া হয়েছিল?" / severe reaction after a previous dose? হ্যাঁ → **YELLOW + refer MO before next dose** (do not auto-schedule).
- Keep the core counselling: **do not restart the series**; give the missed dose and continue. BCG up to 1 y.

Immunization band tally: YELLOW 5–6 (no RED needed — sick-child danger signs route through Child/Infant + pre-sweep). IMM-004 correctly defers vaccination for moderate/severe illness.

7. 🚨 Emergency Case — emergency (Global Emergency Rule Engine)

Module is thin and missing common rural killers in WB. Retain EM-001..005; add:

EM-006 RED HARD-STOP e6 → FRU / DH ≤ 30 min, call 108 [NEW] [DRAFT – SIGN-OFF]
 সাপে কামড়েছে, বিষাক্ত পোকা বা পশুর কামড়?
 Snakebite, venomous sting, or animal bite?
ANSWERS হ্যাঁ · না · নিশ্চিত না
LOGIC IF e6 = হ্যাঁ THEN RED – band locked.
ACTION Immobilise the limb (splint), keep it below heart level, remove tight items, DO NOT cut/suck/tourniquet. Call 108, FRU/DH ≤30 min (anti-snake-venom). Note bite time. Dog/animal bite → also wound wash 15 min + ARV/RIG at facility.
MAPS TO snakebite · scorpion sting · dog bite · rabies risk · envenomation

EM-007 RED HARD-STOP e7 → FRU / DH ≤ 30 min, call 108 [NEW] [DRAFT – SIGN-OFF]
 বিষ/কীটনাশক/অতিরিক্ত ওষুধ খেয়েছে বা খাওয়ানো হয়েছে?
 Poison, pesticide, or overdose ingested?
ANSWERS হ্যাঁ · না · নিশ্চিত না
LOGIC IF e7 = হ্যাঁ THEN RED – band locked.
ACTION Do NOT induce vomiting. Bring the container/label. Protect airway, left-lateral if drowsy. Call 108, FRU/DH ≤30 min.
MAPS TO poisoning · pesticide ingestion · OP poisoning · drug overdose

EM-008 RED HARD-STOP e8 → FRU / DH ≤ 30 min, call 108 [NEW] [DRAFT – SIGN-OFF]

ঠান্ডা-ঘামে ভেজা, খুব দুর্বল নাড়ি, অত্যন্ত নিস্তেজ (শকের লক্ষণ)?

Cold clammy skin, very weak pulse, extreme listlessness (shock)?

ANSWERS হ্যাঁ · না · নিশ্চিত না

LOGIC IF e8 = হ্যাঁ THEN RED – band locked.

ACTION Lay flat, keep warm, raise legs, nothing by mouth. Call 108, FRU/DH ≤30 min.

MAPS TO shock · cold extremities · thready pulse · circulatory collapse

EM-009 RED HARD-STOP e9 → FRU / DH ≤ 30 min, call 108 [NEW] [DRAFT – SIGN-OFF]

গুরুতর আঘাত, বড় ক্ষত থেকে রক্তপাত, বা মারাত্মক পোড়া?

Major injury, bleeding from a large wound, or severe burn?

ANSWERS হ্যাঁ · না · নিশ্চিত না

LOGIC IF e9 = হ্যাঁ THEN RED – band locked.

ACTION Direct pressure on bleeding; for burns cool with clean water, cover, do not apply substances.
Immobilise suspected fractures. Call 108, FRU/DH ≤30 min.

MAPS TO trauma · road accident · major laceration · severe burn · fracture with bleeding

Obstetric-emergency add (confirm placement — Emergency vs ANC) [DRAFT – SIGN-OFF] : in labour with cord visible at the vagina (**cord prolapse**) or labour >12–18 h with no progress (**obstructed labour**) → RED, FRU immediately, knee-chest position for cord prolapse. **Newborn not breathing at birth** → RED + stimulate/clear airway/bag-mask if trained while transport is arranged.

EM-005 re-sweep invariant **retained as-is** — it is the correct global safety lock.

PART C — Master index of additions (for the committee)

Module	New / changed	ID	Band	Why
All paed	NEW	GDS-01..04	RED	IMNCI general danger-sign pre-sweep
Newborn	NEW	NB-008	RED	Hypothermia (PSBI) — was missing
Newborn	NEW	NB-009	RED	Convulsions — was missing
Newborn	NEW	NB-010	RED/Y	Skin pustules / ear pus
Newborn	NEW	NB-011	RED	Bulging fontanelle
Newborn	FIX	NB-006	RED/Y	Split cyanosis vs day-2–14 jaundice
Child	NEW	CH-010	RED	Meningitis (stiff neck)
Child	NEW	CH-011	Y/RED	Dysentery (blood in stool)
Child	NEW	CH-012	Y/RED	Pallor / anaemia
Child	FIX	CH-002/003	—	Make escalations deterministic; dehydration sub-assessment
Infant	NEW	CH-013-i	RED	Lethargy/movement — was missing entirely
Infant	NEW	CH-014-i	RED	Convulsions — was missing
Infant	NEW	CH-015-i	RED	Hypothermia / fontanelle
Infant	FIX	age split	—	0–2 m vs 2–12 m thresholds
Pregnancy	NEW	ANC-008	RED/Y	Severe anaemia — was missing
Pregnancy	NEW	ANC-009	RED/Y	Fever / sepsis — was missing
Pregnancy	NEW	ANC-010	RED	PROM / leaking liquor — was missing
Pregnancy	FIX	ANC-001	RED	Add epigastric pain (HELLP); separate BP/headache
Postpartum	NEW	PNC-007	RED	Postpartum eclampsia — was missing
Postpartum	NEW	PNC-008	RED	PE / DVT / cardiomyopathy
Postpartum	NEW	PNC-009	Y/RED	Postpartum depression screen
Immunisation	FIX	schedule	—	MR not MMR; add PCV, RVV, fIPV×2, Vit-A; WB rollout
Immunisation	NEW	IMM-006	Y	AEFI history
Emergency	NEW	EM-006..009	RED	Snakebite/bite, poisoning, shock, trauma/burns

PART D — Implementation notes (engineering)

These are JSON/engine changes implied by the rules above. I've kept them descriptive rather than shipping merge-ready JSON, because (a) the rules must clear clinical sign-off first, and (b) I don't have your exact

`clinical_decision_engine.json` field schema in front of me.

1. **Answer→band map per rule.** Add an explicit `answer_band_map` to every rule object so intermediate options (খুব কম, কিছুটা, একবার, ...) resolve to a band. Today only `= হ্যাঁ` is honoured. 2. **Uncertainty handling (A2).** Add a `uncertain_policy` field at engine level + a `green_blocked` flag that any `নিশ্চিত না` on a hard-stop sets. GREEN resolution must check `green_blocked == false`. 3. **Escalation sub-questions (A3).** Convert prose escalations into real child questions with their own `condition_set` → band, so the path is deterministic and traceable. 4. **Measurement inputs (A4).** Add numeric-input question types (`type: "numeric"`, `unit`, `thresholds`) gated by a per-cohort `device_available` config. If false, fall back to the yes/no proxy. 5. **Age/GA intake (A5).** Add `age_days` (paeds) and `gestational_weeks` (pregnancy) at case entry; let rules read them in `condition_set`. 6. **Pre-sweep ordering.** GDS-01..04 must evaluate before module rule 1 and obey the same first-hard-stop exit + band-lock + re-sweep invariant (EM-005). Re-sweep must re-check pre-sweep too. 7. **Versioning.** Cut `clinical_decision_engine.json` → v1.1.0 (logic: answer-maps, uncertainty, escalations, pre-sweep) and `triage_cases.json` → v3.0.0 (new questions/rules). Keep a `signoff` block per rule: `{ status: "draft" | "approved", approver, date, protocol_ref }`. The app should be able to **refuse to load any rule whose status ≠ "approved"** in production — so drafts can't ship by accident. 8. **Decision-trace fields.** For MDSR defensibility, log per case: every question, answer, the rule(s) fired, the band, timestamp, engine+cases version, and which signed-off rule version was active.

SIGN-OFF CHECKLIST (AIH&PH / Secretariat clinical committee)

For each [DRAFT – SIGN-OFF] item, the committee confirms: ☒ clinically correct · ☒ correct band · ☒ correct referral target · ☒ Bengali wording is accurate and plain · ☒ within ASHA scope (recognition + referral; any treatment step like MgSO₄/misoprostol is ANM/MO scope unless explicitly delegated).

Specific decisions needed:

- ☐ Uncertainty policy: **Option 1 (force YELLOW)** vs **Option 2 (block GREEN only)** — per PHC capacity.
- ☐ Which **measurement devices** are actually in the WB ASHA kit (enables A4 numeric thresholds).
- ☐ WB-specific **PCV / Rotavirus rollout** status and any district JE endemicity.
- ☐ Exact **fever/temperature and Hb thresholds** for confirmation.
- ☐ PPH "**≥2 pads in 30 min**" proxy (carried over, still open — README item #3).
- ☐ Placement of obstetric emergencies (cord prolapse / obstructed labour / birth asphyxia): Emergency module vs ANC/PNC.
- ☐ Native-Bengali clinical review of all new question strings.

Drafted as a proposed upgrade to the tracked AshaMitra rule files. Bands unchanged: ● Green = home care · ● Yellow = PHC within 24 h · ● Red = FRU/SNCU/DH immediately (≤ 30 –60 min), call 108. A fired RED can never be downgraded (re-sweep invariant). No item here is clinically validated until signed off.

Rendered from AshaMitra_Decision_Rules_v3_DRAFT.md — proposed upgrade to engine v1.0.0 / cases v2.0.0. Bands: ● Green = home care · ● Yellow = PHC within 24 h · ● Red = FRU/SNCU/DH immediately, call 108.